

Software engineering

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Section: B

A Purpose.

- The primary purpose of an E-commerce system is to provide a digital platform that enables buyers to browse, select, and purchase products or services online, while allowing merchant to manage inventory, process payment, and fulfill order efficiently. It provide a 24/7 accessible virtual marketplace without geographic limitations.

B System Users-

• Buyers:-

Browse product catalogs, manage account, place order etc..

• Sellers:-

Lists products, set prices, update inventory, manage order fulfillments etc..

• System Administration:-

Manage user accounts, configure system permission, monitor site performance.

• Customer Support:-

Resolve customer queries, process refund requests, handle disputes.

System Stakeholders.

- Platform owner:
interested in revenue generation, user acquisition, market growth & business scalability.
- Payment Gateway providers:
(e.g. stripe, paypal, Jazz cash etc). Facilitate secure electronic transaction.
- Logistics & Delivery partners:
Depend on accurate order, address and shipping detail to execute physical delivery.
- Tax authority:
Require adherence to consumer protection laws, data privacy regulations.

D Functional Requirements.

- User Authentication:
The system must allow user to register, login, manage profile detail.
- Product catalog:
The system must allow user to search products, apply filter.
- Shopping Cart & Checkout:
The system must support adding / removing items, calculating shipping and taxes and completing checkout via integrated payment methods.
- Order management:
The system must generate unique order IDs, send automated confirmation.

- Inventory management.

The system must generate automatically updated inventory level upon successful sales.

E Non-Functional Requirements.

- Performance.

Web pages and search queries must load within 2 seconds under standard traffic conditions.

- Availability.

The system must maintain 99.9% operational uptime to guarantee continuous access.

- Scalability.

The architecture must horizontally scale to handle high traffic spikes during promotional events.

- Security.

All user credentials and payment transactions must be encrypted using SSL/TLS protocols in compliance with PCI-DSS standards.

- Usability.

The platform must feature a responsive & user-friendly user interface compatible across mobile.

F Potential Risks.

- Data Breach & security Failure.

Unauthorized access to customer personal details or payment credentials leading to financial loss.

- **System Downtime During Peak Sales:-**

Server crashes caused by heavy traffic loads during major sales events, resulting in direct revenue loss.

- **Inventory Misalignment:-**

Discrepancies between digital stock level and actual physical warehouse inventory leading to overselling -

G Quality Attribute:

- **Reliability:**

High consistency in processing payment transactions and maintaining state accuracy without system crashes

- **Security:**

High resilience against common cyber threats (e.g. sql injection, xss) & strict privacy enforcement.

- **Maintainability:**

Modular software design allowing developers to patch bugs, deploy updates -

H Where AI could be used?

- **Personalized product recommendation engine:**

Analyzing user browsing history, search patterns and purchase history to suggest relevant product.

- **AI-Powered Customer Support chatbot:**
Automatically answering common customer inquiries regarding order status, shipping policies & returns 24/7-

- **Dynamics pricing & Demands Forecasting:**
Predicting inventory demand trends & automatically adjusting prices based on market conditions.

I What could Go Wrong if AI is Used Incorrectly:

- **Biased or Discriminatory Pricing:**
AI algorithm applying unfair pricing models or discount based on flawed customer profiling, leading to legal & reputational issues.

- **AI Hallucinations & False commitment:**
AI chatbot providing incorrect product information, non-existent return promises

- **Flawed Demand Forecasting:**
Inaccurate AI stock predictions causing severe overstocking of unwanted items or stockout of popular items